

Model Paper - Database Management Systems

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Q1 (20 marks)

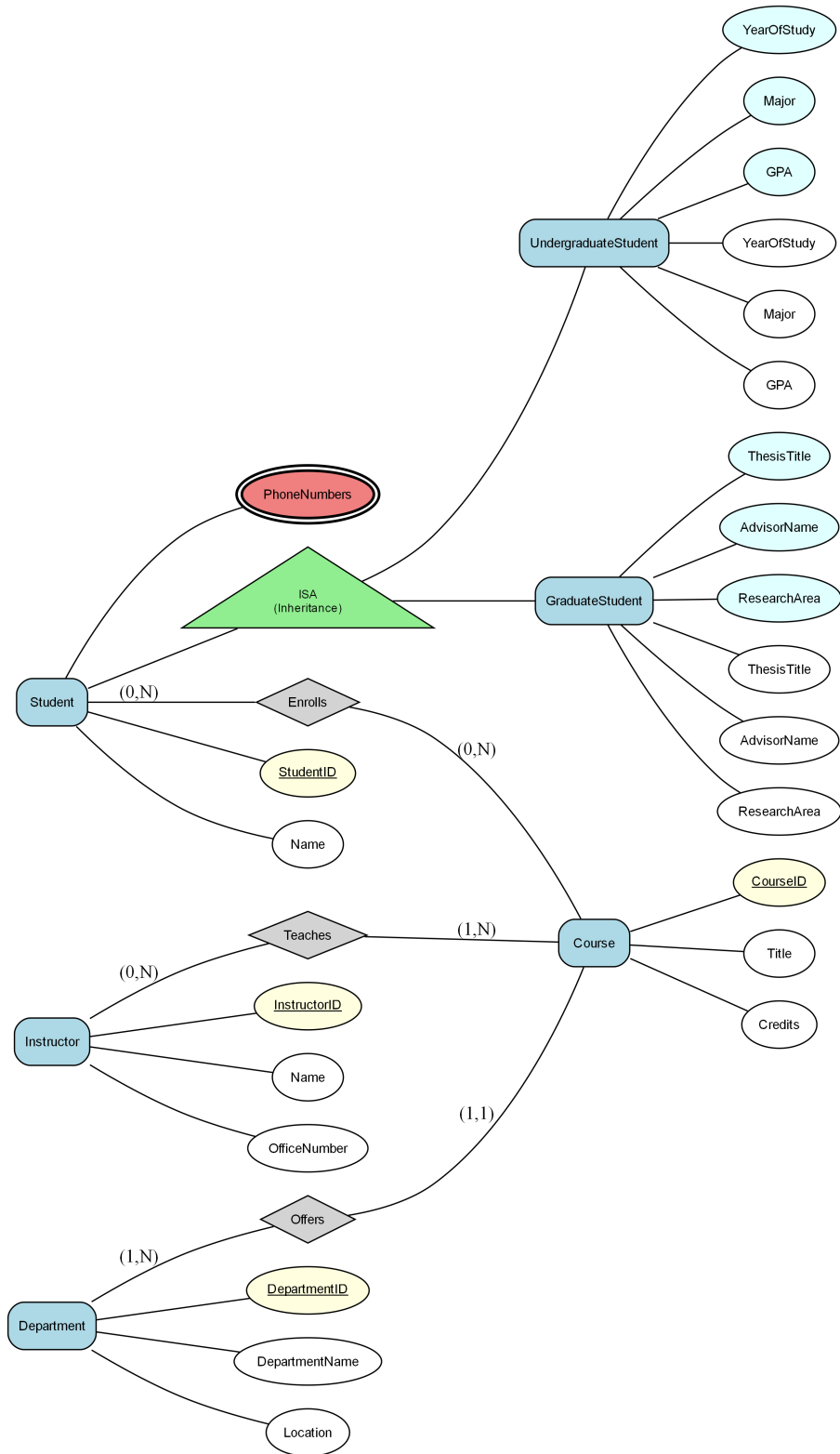
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university database needs to manage its students, courses, instructors, and departments effectively. The Student entity includes StudentID, Name, and PhoneNumbers as attributes. Each Student can enroll in multiple Courses, which has attributes CourseID, Title, and Credits. Instructors, identified by InstructorID, have attributes Name and OfficeNumber, and each Instructor may teach several Courses. The Department entity has attributes such as DepartmentID, DepartmentName, Location., which has DepartmentID and DepartmentName attributes, is responsible for managing the courses and faculty. Each Course belongs to one Department, while each Department offers multiple Courses. Additionally, there is a specialization where certain students can be GraduateStudents with specific attributes such as ThesisTopic and AdvisorName. A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrolls' relationship (Student participation is optional, Course participation is optional). A Instructor can be related to many Courses through the 'Teaches' relationship (Instructor participation is optional). A Department can be related to many Courses through the 'Offers' relationship. Draw the EER diagram for this scenario, representing the relationships and properties clearly

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies over R :

$F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow B, A \rightarrow E\}$. Analyze the structure of this relation to determine its normalization

properties.

a) Find all the keys in relation R using the attribute closure method. (4 marks)

b) Is R in 3NF? Give reasons for your conclusion. (3 marks)

c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A telecommunications company is designing a database to manage its customer subscriptions, service plans, and billing records. The database will store relevant information including customer details, service plan features, and billing history. The initial design involves creating several tables, including Customers, Services, and Billing. Each table will have specific attributes, and certain relationships will be established between them. Based on this scenario, address the following questions.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment: ````sql CREATE TABLE Students (StudentsID INT PRIMARY KEY IDENTITY(1,1), Name VARCHAR(50) NOT NULL, Email VARCHAR(100) UNIQUE, Age INT CHECK (Age >= 18), Status VARCHAR(20) DEFAULT 'Active'); INSERT INTO Students (Name, Email, Age) VALUES ('John Smith', 'john@example.com', 25); ```` In the context of the scenario, identify the type of constraints that should be applied to enforce data integrity. Briefly explain how each constraint will be applied. (2 marks)

c) Code Segment: ````java String sql = "SELECT * FROM employees WHERE department = ?"; PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs = pstmt.executeQuery(); ```` Which type of JDBC statement is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

d) Briefly explain the options available for handling NULL values in SQL tables, particularly within the context of this telecommunications database. (3 marks)

e) An organization is establishing role-based access control for its database system. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the operational database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating transaction data.

i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)

ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)

iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the operational database using a user-defined server role. (5 marks)

iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)

v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the operational databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a Hospital: Patient (patientId: int, name: varchar(100), address: varchar(150), dob: date) Doctor (doctorId: int, name: varchar(100), specialty: varchar(50), phone: varchar(15)) Appointment (appointmentId: int, patientId: int, doctorId: int, appointmentDate: date, status: varchar(20)) Payment (paymentId: int, appointmentId: int, paymentAmount: real, paymentStatus: varchar(20)) The 'Patient' table stores information about patients including unique patient ID and personal details. The 'Doctor' table contains details about doctors and their specialties. The 'Appointment' table keeps track of all appointments scheduled between patients and doctors, while the 'Payment' table records payment information related to these appointments. The 'Appointment' table manages appointment records, tracking scheduled appointments between patients and doctors with dates and times. The 'Payment' table contains relevant information for the database system.

a) Write SQL Queries to perform the following: i. Find the name and phone number of the doctor who has an appointment scheduled for '2023-12-01'.

i. Find the name and phone number of the doctor who has an appointment scheduled for '2023-12-01'. (4 marks)

ii. Find the patient who has the highest total payment amount compared to all other patients. Find the patientId and name. (6 marks)

iii. Find the name and name from Doctor, Patient, and Appointment where Appointment.appointmentDate is not null. (7 marks)

b) b) Create a function to calculate the total payment amount for a given patient. This function should account for payment with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'Payment' column in the 'Payment' table whenever a new payment record is added, or an existing payment record is updated. The 'Payment' column should contain the total payment amount for each patient. Ensure the trigger adjusts the 'Payment' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total payment amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)